

CHAPTER 1

INTRODUCTION

1.1 Project Context

Zamboanga City is rapidly positioning itself as a major hub for cacao production and chocolate manufacturing in the Zamboanga Peninsula. According to the Department of Agriculture, the quality of fermented cacao beans is the single most important factor in determining their market price and export potential. High-quality beans require precise fermentation and drying processes to achieve the chemical profiles necessary for premium chocolate production. For the local industry to thrive and compete globally, ensuring that only the highest grade of beans reaches the market through an affordable YOLOv8-based automated sorting system is an essential economic requirement for Zamboanga cooperatives and smallholder farmers.

However, the majority of local cacao farmers and small-scale cooperatives in Zamboanga still rely on manual sorting and subjective visual inspection to grade their beans. This manual process is slow, inconsistent, and highly prone to human error, where defects such as mold, shriveled beans, or uneven fermentation are frequently missed. Research in agricultural logistics shows that if even a small percentage of low-quality or moldy beans is mixed into an export-grade batch, the entire shipment can be devalued or rejected by international buyers. This results in significant financial losses for Zamboanga farmers who lack access to professional, automated grading equipment at the local cooperative level.

Current technological solutions in the local cacao industry are mostly limited to basic moisture meters, which only measure one specific aspect of bean quality. A significant research gap exists because existing software-only solutions through a mobile screen do not provide a complete physical sorting solution. Furthermore, existing industrial color sorters are prohibitively expensive and too large for small- to medium-scale Zamboanga cooperatives to acquire. There is a clear need for an integrated, low-cost automated machine that can perform both variety identification and physical grading-based sorting autonomously without requiring constant human labor at barangay-level processing facilities.

The research niche of this study lies in cooperative-scale post-harvest analysis for the three Philippine cultivar groups cultivated nationwide: Criollo, Forastero, and Trinitario. Farmers in Barangay Ayala and Pasonanca still perform repetitive hand-sorting for hours, separating beans that carry different market values, yet no affordable validated system in Zamboanga currently integrates cloud-hosted dual-model YOLOv8 inference with ESP32-controlled mechanical sorting. Parallel advances in staged machine vision, RTSP industrial cameras, and accessible IoT microcontrollers have created a viable pathway to bridge the gap between fully manual sorting and industrial automation.

To address these issues, this study proposes CacaoScan, an AIoT sorting machine designed to automate the post-harvest quality control cycle. The Capstone 1 prototype uses a conveyor-fed staging pipeline that queues beans in fixed batches of three, captures a controlled vision image inside a light-isolated tower, and sends that image to a cloud-hosted dual YOLOv8 pipeline for per-bean variety and quality classification. An ESP32-S3 controller then executes a sequenced discharge cycle in which lateral pushers and a high-torque sorting flipper divert each classified bean into Export-Grade, Needs Drying, or Rejected bins. The system is designed to operate in standalone mode at the client site and is managed through a React Native mobile application and a responsive web dashboard aligned with SDG 9 and SDG 12.

1.2 Statement of the Problem

The cacao post-harvest industry in Zamboanga City faces a systemic challenge rooted in high labor cost, subjectivity, and inaccuracy of manual cacao bean sorting, which leads to batch rejection when a few defective beans lower the market value of an entire harvest. Cooperative workers perform repetitive visual inspection for extended periods, yet human fatigue and inconsistent judgment allow mold, shriveled beans, slaty coloration, and uneven fermentation to pass undetected into export-bound batches. Export buyers and domestic processors increasingly expect grading decisions to reflect documented quality criteria rather than individual estimator preference, yet manual workflows in barangay facilities rarely produce repeatable results across shifts or between workers. These conditions create a structural barrier to the region's ability to consistently deliver export-ready cacao at cooperative scale.

The inability to segregate Criollo, Forastero, and Trinitario beans accurately during sorting further reduces average selling prices when high-value beans are blended with lower-value cultivars in Barangay Ayala, Pasonanca, and surrounding communities. Variety mixing dilutes the market identity of batches that could otherwise command premium pricing in specialty and single-origin channels. At the same time, visual quality decisions are supposed to align with the Philippine National Standard for Cacao (PNS/BAFS 58:2019), which defines export-relevant defect thresholds for mold, slaty beans, insect damage, and foreign matter, but manual inspection cannot apply those criteria with uniform consistency during long sorting sessions. Cooperative managers therefore struggle to defend grading outcomes when buyers question batch homogeneity or defect levels.

A further operational problem is the absence of machine-verifiable documentation for how each post-harvest batch was classified before shipment or cooperative resale. Manual sorting produces no persistent per-bean or per-batch record that links grading decisions to timestamps, operator actions, or defect categories. When a batch is questioned or rejected, cooperatives cannot quickly determine whether the error originated in farm handling, post-harvest drying, or the sorting table itself. This documentation gap weakens trust with buyers and slows corrective action in cooperative quality programs.

Compounding these problems is the absence of an affordable technological solution that bridges screen-only assessment tools and industrial-grade automation. Existing mobile and web-based cacao assessment tools detect defects visually but cannot physically separate beans by variety or quality grade, leaving the most labor-intensive step unresolved. Industrial color sorters remain economically inaccessible to smallholder cooperatives, while no locally developed standalone AIoT system in the Zamboanga Peninsula currently integrates cloud-hosted dual-model YOLOv8 inference, infrared-synchronized batch staging, and cooperative-scale monitoring through both mobile and web platforms. This study addresses that documented gap through the design and development of CacaoScan.

1.3 Purpose and Description

The primary function of CacaoScan is to automate the classification and physical sorting of dried fermented cacao beans during the post-harvest processing stage, replacing the manual one-by-one inspection currently performed by cooperative workers in Zamboanga City. The Capstone 1 prototype receives beans through a funnel hopper and V-trough conveyor that aligns oblong beans lengthwise before an infrared beam-break sensor counts a fixed staging batch of three beans at a servo-controlled gate. Once the batch is staged, a tilting distributor chute deposits the beans into three isolated pockets inside a light-controlled vision tower, where they remain stationary for imaging and cloud analysis before sequential mechanical discharge. This staged mechanical presentation is designed to give the vision system a repeatable view of each bean position while remaining buildable within the cost and fabrication constraints of a cooperative-scale capstone prototype.

The operational sequence continues when an Industrial HD IP Camera mounted on the vision tower captures a single high-resolution image of all three pocketed beans and transmits it via RTSP over cooperative Wi-Fi to a cloud-hosted FastAPI service on Hugging Face Inference Endpoints. Model A classifies the variety of each bean by pocket position as Criollo, Forastero, or Trinitario, while Model B assigns a quality grade as Export-Grade, Needs Drying, or Rejected using visible indicators of mold, shriveling, and slaty coloration aligned with PNS/BAFS 58:2019 visual grading practice. The Node.js backend receives the fused inference result for each pocket and forwards the corresponding discharge command to the ESP32-S3 firmware without requiring an on-site inference laptop at the cooperative facility. Cloud deployment keeps heavy YOLOv8 processing off the prototype chassis while still allowing the machine to operate as a standalone client-site unit.

After inference is complete, the ESP32-S3 executes a sequenced mechanical discharge cycle for each bean in the staged batch. The MG996R sorting flipper first pre-positions to the assigned bin path, an SG90 lateral pusher ejects the bean from its pocket onto a common sorting rail, and the firmware waits approximately 800 milliseconds for the rail to clear before releasing the next bean. Power and control electronics are separated through dual LM2596 buck converters, an opto-isolated relay for the conveyor motor, and fuse-protected twelve-volt distribution so actuator loads do not reset the microcontroller during sorting. This arrangement is

intended to improve mechanical reliability during repeated batch cycles in a barangay processing environment.

Beyond the physical sorting mechanism, CacaoScan incorporates a software layer designed to provide real-time operational visibility and permanent quality documentation for every processed batch. Each per-bean classification result is logged to Supabase through the Node.js backend API, enabling the React Native mobile application to display live batch statistics and the responsive web dashboard built with React JS and Tailwind CSS to present quality trends, PDF certificate generation, and management analytics. Cloudinary archives tower images for traceability, while Firebase Cloud Messaging delivers push alerts for batch completion and quality threshold events. Together, these services convert each sorting session into a reviewable digital record that cooperatives can use for internal quality control and buyer communication.

What makes CacaoScan unique, innovative, and relevant is its integration of conveyor-fed batch staging, enclosed industrial vision, cloud-hosted dual-model inference, and sequenced multi-actuator discharge into one affordable machine designed for barangay-level cacao cooperatives in the Zamboanga Peninsula. Unlike screen-only tools, CacaoScan completes the full grading-and-sorting cycle autonomously at the client site without requiring an on-site laptop for inference. Unlike industrial sorters, it is built from accessible components including an ESP32-S3 controller, SG90 and MG996R servos, opto-isolated relay switching, and dual LM2596 power rails, making it financially viable for smallholder facilities while remaining expandable toward higher-throughput singulation in later development phases. Performance targets for the prototype evaluation include detection accuracy above 85 percent, mechanical sorting success rate of 95 percent, reliable completion of batch-of-three vision and discharge cycles, and a UAT weighted mean above 4.20 on the Likert scale.

1.4 Objectives

The main objective of this capstone project is to design, develop, and evaluate CacaoScan, an AIoT-driven machine capable of automated cacao bean variety classification and quality-based physical sorting using standalone industrial vision and dual-model YOLOv8 computer vision. This objective encompasses the infeed and conveyor transport module, the batch staging and vision tower module, the cloud-hosted AI inference pipeline, the ESP32-S3 multi-actuator discharge system, the isolated control electronics enclosure, and the React Native and responsive web management interfaces. The study aims to demonstrate measurable improvements over manual expert grading using confusion matrix analysis, mechanical sorting success rate, batch-cycle reliability, and User Acceptance Testing rather than qualitative consistency claims alone.

A further objective is to produce a validated dual-model YOLOv8 training and deployment pipeline calibrated to Zamboanga bean varieties and defect profiles, using a hybrid dataset of 3,000 to 5,000 annotated images that includes at least 500 locally captured samples

from cacao farms in Barangay Ayala and Pasonanca. Models optimized through ONNX and TensorRT quantization will be served through Hugging Face Inference Endpoints to classify each bean position within staged tower images while the physical machine operates autonomously at the cooperative facility. Success is evaluated against detection accuracy above 85 percent, mechanical sorting success rate of 95 percent, and a UAT weighted mean above 4.20 on the Likert scale.

1.4.1 Specific Objectives

1. To design and construct an infeed and transport module consisting of a funnel hopper, vibratory feed assist, and V-trough conveyor system that delivers aligned cacao beans to a staging gate at a controlled cooperative-scale feed rate, by June 2027.
2. To implement a batch staging module using an infrared beam-break sensor and servo-controlled gate that forms fixed batches of three beans before distribution into the vision tower, by June 2027.
3. To develop an enclosed vision tower with an Industrial HD IP Camera, LED ring lighting, and a three-pocket imaging tray that supports stable multi-bean capture for cloud inference, by June 2027.
4. To develop a dual-model YOLOv8 cloud pipeline in which Model A identifies Criollo, Forastero, and Trinitario varieties and Model B detects mold, shriveling, and slaty coloration for each pocket position using PNS/BAFS 58:2019-aligned visual grading criteria, achieving detection accuracy above 85 percent, by June 2027.
5. To implement an ESP32-S3 discharge and sorting module using a tilting distributor chute, SG90 lateral pushers, a common sorting rail, and an MG996R three-way sorting flipper that routes each bean to Export-Grade, Needs Drying, or Rejected bins, by June 2027.
6. To integrate isolated control electronics including dual LM2596 buck converters, an opto-isolated conveyor relay, fuse protection, and a React Native mobile application with a responsive web dashboard for monitoring and manual override, by June 2027.
7. To evaluate system performance using a 150-bean expert-labeled test set, mechanical sorting success rate measurement, batch-cycle timing analysis, and User Acceptance Testing with five cooperative farmers, targeting a Likert weighted mean above 4.20, by June 2027.

1.5 Scope and Limitation

The scope of this study encompasses the complete design, development, and cooperative-scale prototype evaluation of CacaoScan as an integrated standalone AIoT machine for staged-batch cacao classification and physical sorting. In terms of hardware, the scope covers the funnel hopper and V-trough conveyor infeed, the infrared-counted staging gate and tilting

distributor, the enclosed vision tower with three-pocket tray and Industrial HD IP Camera, the lateral pusher array and MG996R sorting flipper with three output bins, and the transparent IoT control box containing an ESP32-S3, dual LM2596 buck converters, opto-isolated relay, and fuse-protected 12V distribution. In terms of software, the scope includes dual YOLOv8 model training on an NVIDIA GPU workstation, cloud deployment through Hugging Face Inference Endpoints with FastAPI, Node.js and Express.js backend synchronization, Supabase database logging, Cloudinary image archival, Firebase Cloud Messaging notifications, React Native mobile control, and a React JS responsive web dashboard. Evaluation is limited to the Capstone 1 prototype configuration described in this proposal rather than to a commercially certified inspection instrument.

Supported cacao varieties are limited to Criollo, Forastero, and Trinitario for Model A classification. Supported visual defect categories for Model B include mold, shriveled beans, and slaty coloration interpreted against the export-preparation intent of PNS/BAFS 58:2019, which provides the national reference for cacao bean grading and classification in the Philippines. Physical sorting outputs are limited to three bins labeled Export-Grade, Needs Drying, and Rejected, corresponding to the cooperative post-harvest categories used in the project prototype. The Capstone 1 machine processes dried fermented cacao beans in fixed batches of three per vision cycle, with sequential discharge controlled by timed flipper positioning and approximately 800 milliseconds of rail-clearance delay between beans.

The dataset strategy uses a hybrid approach sourcing base images from Kaggle and Roboflow, supplemented by at least 500 original photos from local cacao farms in Barangay Ayala and Pasonanca for Zamboanga-specific calibration, with a total annotation target of 3,000 to 5,000 images labeled in Roboflow at 640 by 640 pixels with augmentation and normalization preprocessing. Annotations support both single-bean and multi-pocket tower images so Model A and Model B can return per-position variety and grade labels during staged vision cycles. Ground-truth validation will involve certified cacao graders or DA and BFAR experts who compare machine-assigned grades with manual expert sorting during the 150-bean controlled test set. Confusion matrix analysis, mechanical sorting success rate, and User Acceptance Testing with five cooperative farmers will be used to determine whether the prototype meets the study's performance thresholds.

The limitations of this study are defined by the boundaries of the prototype's intended use and by the constraints of a capstone development timeline. The Capstone 1 machine uses batch-of-three staging rather than continuous single-bean singulation, which simplifies mechanical control but limits immediate throughput compared with industrial continuous sorters. The system does not process wet or semi-fermented beans, does not perform cut-test internal inspection, and treats variety output as a cooperative sorting aid rather than laboratory genetic certification. Cloud inference requires stable Wi-Fi at the cooperative site during operation, and CacaoScan is developed as an experimental research prototype rather than a commercially

certified agricultural inspection device without further validation, field deployment, and regulatory review beyond the scope of this study.

1.6 Significance of Study

The theoretical significance of this study lies in its contribution to documented frameworks for standalone AIoT agricultural sorting systems that combine cloud-hosted dual-model inference with staged mechanical vision and sequenced multi-actuator discharge. CacaoScan demonstrates how enclosed tower imaging, Hugging Face-hosted YOLOv8 models, and ESP32-synchronized servo control can be integrated into one cooperative-scale pipeline with measurable precision, recall, F1-score, and mechanical reliability metrics reported for each classified bean position. The study further shows how a batch-staging compromise can be used in early prototype research without abandoning per-bean classification as the unit of analysis for model evaluation. This contribution responds to the need for academically reportable AIoT architectures that separate cloud inference from edge actuation while still producing traceable physical outcomes.

The practical significance of CacaoScan is demonstrated through its direct application to cacao cooperatives in Zamboanga City, where manual sorting remains dominant despite rising export quality demands. By automating variety segregation and defect-based grading within a buildable conveyor-and-tower prototype, the system helps farmers reduce labor burden, improve batch consistency, and minimize rejection risk during cooperative post-harvest operations. Quality decisions grounded in PNS/BAFS 58:2019 visual criteria can be applied more uniformly than table-side estimation alone, while digital logs stored in Supabase and exportable PDF certificates from the web dashboard provide cooperatives with verifiable batch records for buyers and internal audits. The prototype therefore offers a realistic intermediate technology between manual sorting tables and industrial automation that barangay facilities can actually afford to build and test.

The social significance of this study extends to farming households and cooperative workers who perform repetitive sorting during peak harvest seasons in the Zamboanga Peninsula. Automating the inspection and discharge stages reduces occupational strain associated with long manual sorting shifts and lowers dependence on a small number of experienced estimators whose absence can disrupt processing schedules. Making export-oriented grading support accessible at cooperative scale can also strengthen the income stability of smallholder households that depend on cacao as a cash crop. User Acceptance Testing with five cacao farmers from local cooperatives ensures that the mobile and web interfaces reflect real operational needs at the client site rather than laboratory assumptions alone.

The academic significance of this study is reflected in its value as an end-to-end IT capstone reference integrating mechanical feed design, computer vision, cloud inference deployment, IoT firmware, power-isolated electronics, and full-stack monitoring within one WMSU capstone project. The documented staging workflow, training pipeline, ONNX and

TensorRT optimization path, and evaluation methodology provide a replicable framework for future AIoT agricultural automation research in Western Mindanao. Students and faculty reviewers can use the project as a case study in how capstone teams translate a complex machine concept into a testable prototype with separate software, hardware, and documentation deliverables. The resulting chapter structure, metrics plan, and prototype traceability also model how client-based cooperative projects can be reported with academic rigor at the proposal stage.

1.7 Definition of Terms

Defined terms are those whose meanings in the study are particular or unique. The following terms are operationally defined as they are used throughout this research.

Table 1. Definition of Terms

Term	Definition
1. AIoT (Artificial Intelligence of Things)	The convergence of Artificial Intelligence and Internet of Things technologies, where AI algorithms integrated with IoT devices enable autonomous decision-making from sensor and vision data (Shi et al., 2016). In CacaoScan, AIoT refers to cloud YOLOv8 inference linked to ESP32-controlled mechanical sorting.
2. YOLOv8 (You Only Look Once version 8)	The eighth generation YOLO real-time object detection architecture developed by Ultralytics for simultaneous bounding-box prediction and class classification (Jocher et al., 2023).
3. RTSP (Real Time Streaming Protocol)	A network protocol used to stream vision data from the Industrial HD IP Camera to the cloud inference service during tower image capture in CacaoScan.
4. Dual-Model AI Pipeline	An architecture in which Model A performs variety classification and Model B performs quality grading on each bean position within a staged vision image before discharge logic determines the output bin.

5. Hugging Face Inference Endpoints	A cloud hosting service used to deploy optimized YOLOv8 models for inference accessible to the standalone machine over Wi-Fi.
6. Batch-of-Three Staging	A prototype control strategy in which exactly three beans are counted, stopped, imaged, classified, and discharged as one processing cycle before the next conveyor batch begins.
7. Vision Tower	An enclosed, light-isolated imaging chamber with a three-pocket tray, LED ring lighting, and overhead Industrial HD IP Camera used to capture stable multi-bean images.
8. V-Trough Conveyor	A conveyor with angled side walls that encourages cacao beans to align lengthwise during transport from the hopper to the staging gate.
9. Staging Gate	A servo-controlled stop point with an infrared beam-break sensor that counts three beans and pauses the conveyor motor before distribution into the vision tower.
10. Lateral Pusher Array	A set of SG90 micro-servos mounted outside the vision tower that push classified beans from their pockets onto the common sorting rail.
11. Common Sorting Rail	A shared inclined slide that guides discharged beans from the vision tower toward the sorting flipper and output bins.
12. MG996R Sorting Flipper	A high-torque metal gear servo that pre-positions a flap to route each bean into Export-Grade, Needs Drying, or Rejected bins.

13. ESP32-S3 Microcontroller	The Wi-Fi-enabled master controller that manages conveyor relay switching, batch counting, cloud command handling, and servo sequencing.
14. Opto-Isolated Relay	An electrical isolation component that switches the 12V conveyor motor without feeding motor noise back into the ESP32 logic circuit.
15. React Native	A cross-platform mobile framework used for field machine control, batch monitoring, and operator notifications.
16. Responsive Web Dashboard	A React JS and Tailwind CSS management interface for quality analytics, trend visualization, and PDF certificate generation.
17. Supabase	A PostgreSQL-based backend service for real-time batch logs, user profiles, and quality record persistence.
18. Confusion Matrix	An evaluation table reporting true and false classifications per class, used to compute precision, recall, and F1-score.
19. User Acceptance Testing (UAT)	End-user evaluation of the mobile and web interfaces by five cacao farmers from local cooperatives using a Likert-scale survey.
20. Weighted Mean	A statistical measure used to interpret Likert-scale UAT responses; a mean above 4.20 indicates high acceptability.

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